Energy Reports 15 (2026) 108846

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Electricity load forecasting using hybrid prophet–TFT model incorporating weather data

Nazrul Amin [a], Yong-Woon Kim [a], Chulung Kang [b], Yung-Cheol Byun [c,*]

[a] Department of Computer Engineering, Jeju National University, Jeju 63243, Republic of Korea
[b] Department of Mechanical Systems Engineering, Jeju National University, Jeju-si 63243, Republic of Korea
[c] Department of Computer Engineering, Major of Electronic Engineering, Food Tech Center (FTC), Jeju National University, Jeju 63243, Republic of Korea

ARTICLE INFO

Keywords:
Electricity load forecasting
Hybrid model
Temporal Fusion Transformer (TFT)
Weather data integration
Energy management
Feature importance

ABSTRACT

Accurate forecasting of electrical consumption is essential for controlling the integration of renewable energy sources and improving the electric grid's performance. We introduce a new hybrid forecasting method in this study that combines Facebook Prophet's time-series decomposition with the deep learning capabilities of a Temporal Fusion Transformer (TFT). By leveraging Prophet to model long-term trends and seasonality and using TFT to capture short-term patterns and complex multivariate relationships, the hybrid model effectively bridges statistical and neural forecasting techniques. We incorporate real-time meteorological features from Jeju Island (using data from 2012–2020) to enhance forecast adaptability to weather variations. The proposed Prophet–TFT model achieves high accuracy, outperforming each individual model and other benchmark hybrids. It significantly improves upon baseline predictions, achieving an MAE of 8.4 MW with a MAPE of about 1.43%. The inclusion of weather data, particularly during extreme temperature conditions, further enhances forecasting precision. An attention-based feature importance analysis reveals that key meteorological variables, such as Jeju's air temperature, Seogwipo's wind speed, and a heat-discomfort index, are among the most influential predictors. These results highlight the significant role of environmental conditions in shaping electricity consumption. The results illustrate the remarkable efficacy of the hybrid model, interpretable and flexible. It offers a practical tool for grid operators, especially in regions with significant renewable resources and weather-dependent demand, to reliably forecast loads and enhance the stability of power systems.

1. Introduction

Accurate electricity load forecasting is essential for power system planning and stability (Yazici et al., 2022). To ensure that supply meets demand, precise forecasts help balance the grid and prevent outages, leading to greater efficiency and reduced operating costs (Bassey, 2023). These capabilities are increasingly important as power systems evolve with high penetration of intermittent renewables and extreme weather events. For example, Jeju Island experiences pronounced seasonal demand swings due to hot summers and cold winters, making reliable forecasts there critical for grid resilience. Traditionally, statistical models such as ARIMA and SARIMA have been used to capture linear trends and periodic effects via differencing, autoregression, and seasonal components (Szostek et al., 2024); they remain attractive for their parsimony and interpretability but often fail to model nonlinear dynamics, interaction effects, or sudden regime shifts in demand (ArunKumar et al., 2022). Generalized Additive Models (GAMs) extend this classical toolbox by learning smooth, interpretable nonlinear effects (e.g., temperature curves, time-of-day, and holiday splines), which improves over purely linear specifications (Khamma et al., 2020); however, GAMs struggle as covariate dimensionality and cross-feature interactions grow, and they do not natively support multi-horizon quantile forecasting or long-range temporal dependencies. Simple machine-learning baselines such as Feedforward Neural Networks (FFNNs) and tree/boosting methods (e.g., XGBoost) capture generic nonlinearities but rely on hand-crafted lag features and calendar encodings; without an explicit temporal inductive bias, they can underperform on sequence structure and provide limited calibrated uncertainty (Mansoor et al., 2021; Nielsen, 2016).

In recent years, deep learning techniques (CNNs, LSTMs, and Transformers) have gained popularity for load forecasting (Rafi et al., 2021), due to their ability to learn complex temporal relationships (Aslam et al., 2021). In a precise sense, this includes simultaneously modeling multi-scale seasonality (hourly–weekly–annual), long-range dependencies, holiday/calendar effects, and nonlinear weather–load interactions via convolution, gating, and attention mechanisms (Qiu et al., 2023).

* Corresponding author.
E-mail addresses: nazrulamin@stu.jejunu.ac.kr (N. Amin), ywkim@jejunu.ac.kr (Y.-W. Kim), cukang@jejunu.ac.kr (C. Kang), ycb@jejunu.ac.kr (Y.-C. Byun).

https://doi.org/10.1016/j.egyr.2025.11.105
Received 25 May 2025; Received in revised form 11 November 2025; Accepted 29 November 2025
Available online 6 January 2026
2352-4847/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

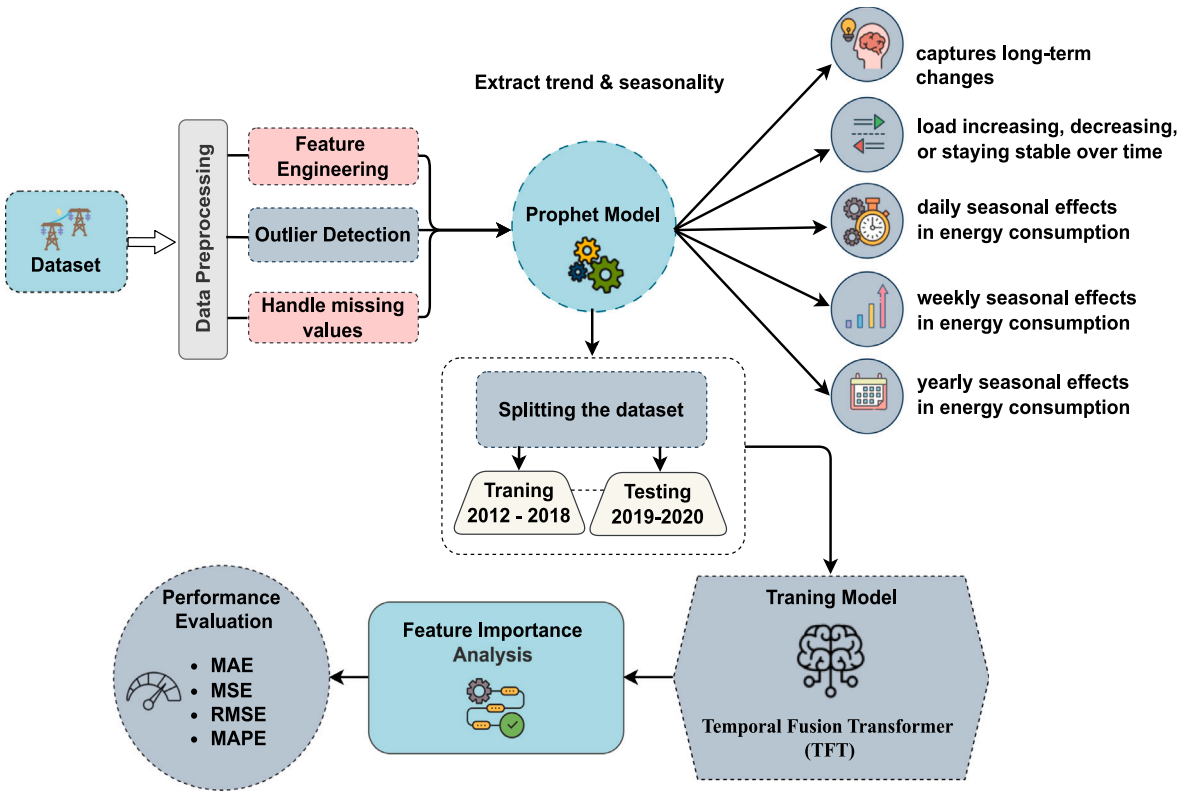


Fig. 1. Overview of the proposed methodology. The figure illustrates the full forecasting pipeline, including data preprocessing, Prophet-based trend and seasonality extraction, train-test splitting, TFT training, and evaluation with feature importance analysis.

Many recent hybrid models improve over single families but still struggle to cover all regimes simultaneously: CNN–LSTM learns local motifs but lacks an explicit calendar prior (Wan et al., 2023) XGBoost–LSTM relies on hand-crafted lags and offers weak native uncertainty; GAMs provide interpretable smooth trends but are less effective with high-dimensional covariates and multi-horizon uncertainty; EMD–BiLSTM denoises via signal decomposition and captures bidirectional dependencies, but suffers from recomposition error and higher complexity.; and Prophet–LSTM (Bashir et al., 2022) inherits Prophet's interpretability but lacks horizon-wise variable selection and attention. We address these gaps with a Prophet–TFT hybrid. The backbone (Prophet) yields an interpretable, parametric decomposition piecewise trend with sparse changepoints, weekly/annual Fourier seasonalities, and country-specific holiday effects producing a clean residual. A Temporal Fusion Transformer (TFT) then models that residual with variable-selection networks that gate static, known-future, and observed-past covariates at each time step; gated LSTM encoder–decoder blocks for short/medium-term dynamics; multi-head attention to fuse the most relevant past when issuing each horizon; and a quantile output layer for calibrated forecasts. This division of labor, Prophet for low-frequency and holiday structure, TFT for nonlinear interactions and spikes, yields both interpretability (from Prophet components) and calibration from TFT's selection, gating, and attention. The resulting transparency facilitates validation, error diagnosis, what-if analysis, and compliance by exposing the inputs and time periods that influence a forecast, while maintaining state-of-the-art accuracy.

In our framework, Prophet is used as a feature extractor: its fitted trend and seasonal components are fed as inputs to the TFT, alongside time and weather covariates. This way, the TFT's gating and attention mechanisms can focus on learning short-term and nonlinear dynamics from both the decomposition-based features and the raw meteorological data. The complete forecasting framework, from data preprocessing to model evaluation, is illustrated in Fig. 1. In other words, Prophet provides structured, interpretable components (trend, multiple seasonality) (Toharudin et al., 2023), and the TFT refines these with a deep nonlinear model. Prophet was chosen over other decomposition methods because it natively handles multiple seasonalities and provides built-in changepoint detection. Where generic methods like STL or EEMD (Song et al., 2022b) either require manual parameter tuning or struggle under noisy, high-frequency fluctuations. Thus, the Prophet–TFT synergy offers a complementary blend of statistical interpretability and deep temporal learning (Zhou et al., 2024).

We evaluate our model on Jeju Island's hourly load data (2012–2020) with multistation weather inputs (temperature, humidity, wind, pressure, etc.). The hybrid approach achieves better results compared to standalone statistical or hybrid neural models. Key meteorological features, such as air temperature, wind speed, and a heat-discomfort index (a combined temperature-humidity metric), from stations across Jeju are used as covariates along with time of day and seasonal factors. Including these weather features enables the model to account for demand variability under extreme weather conditions. The proposed hybrid model not only improves forecasting performance but also enhances explainability. Our key contributions are as follows:

- **Hybrid Prophet–TFT forecasting framework:** We introduce a novel load forecasting model that combines Prophet's trend/seasonality decomposition with TFT's attention-based deep learning. This hybrid significantly improves results over existing models.
- **Dynamic weather integration:** The model incorporates real-time, multistation meteorological data into both the Prophet decomposition and the TFT network, allowing forecasts to adapt dynamically to weather variations.
- **Comprehensive performance evaluation:** We validate the approach on Jeju Island's nine-year dataset (2012–2020). The hybrid model consistently outperforms traditional statistical methods and single-model baselines, achieving a MAPE of 1.43%.

- **Feature impact analysis:** We identify and quantify the most influential input variables using attention-based importance from the TFT. This shows how well the model can convey and explain key seasonal and weather-driven effects.

By addressing current limitations in STLF methodologies, our hybrid Prophet–TFT model delivers a more accurate, adaptable, and interpretable forecasting solution. This enhanced forecasting capability is crucial for enabling effective energy management and optimizing grid operations, especially in regions characterized by significant seasonal and weather-induced demand fluctuations, such as Jeju Island.

2. Related work

Accurate forecasts enhance security of supply, reduce operating costs, and support renewable integration across timescales. Electricity load forecasting is commonly scoped by horizon because the dominant drivers, data availability, and decision stakes change with lead time (Nti et al., 2020). We distinguish (i) *short-term* (minutes–day ahead), (ii) *mid-term* (week–month ahead), and (iii) *long-term* (seasonal–multi-year) settings (Hammad et al., 2020).

short-term (minutes–day ahead): This horizon supports real-time and day-ahead operations such as unit commitment, dispatch, reserves, and demand response. Load is shaped by pronounced diurnal/weekly regularities, high-frequency meteorology, and holiday or event effects; very recent history is especially informative (Zhang et al., 2018). Best practice evaluates with rolling-origin, expanding-window backtests over 1–24 h horizons and reports both point and probabilistic accuracy, reflecting the need to control imbalance costs and operational risk. This study primarily targets the short-term horizon.

Mid-term (week–month ahead): Mid-term forecasts inform maintenance and outage scheduling, fuel and hydro planning, tariff and workforce planning, and storage policies (Kuster et al., 2017). Robustness to seasonal transitions and medium-range weather outlooks is essential, and models typically incorporate stronger priors and explicit calendar structure. Evaluation commonly relies on blocked or seasonal cross-validation, with emphasis on stable performance and credible uncertainty bands to support planning decisions (Ziel and Steinert, 2018).

Long-term (seasonal–multi-year): Long-term forecasts guide capacity expansion, policy and electrification strategies, renewable integration, and network upgrades (Aditya et al., 2025). Uncertainty is dominated by macro trends, technology adoption, and structural change; the focus shifts from short-run accuracy to scenario consistency, stress testing, and propagation of uncertainty across temporal and spatial scales. Although not the focus here, long-horizon insights set boundary conditions for short- and mid-term operations (Azeem et al., 2021). An overview of representative methods and their limitations is summarized in Table 1.

2.1. Classical statistical models

Classical statistical models remain important baselines in electricity load forecasting due to their interpretability and efficiency (Kuster et al., 2017). ARIMA has been widely applied but often struggles with nonlinear consumption patterns and weather-driven variability. SARIMA extends ARIMA to capture seasonality and performs well on daily or weekly cycles, though its accuracy declines in highly volatile or irregular demand settings (Tarmanini et al., 2023; Valipour, 2015). Other approaches, such as Holt–Winters/ETS and GAMs, provide flexibility for trend and nonlinear covariates but remain limited when handling high-dimensional or rapidly changing data (Koehler et al., 2001; Mathivha et al., 2020). More recently, Prophet has gained popularity for its automated decomposition of trend, seasonality, and holidays, offering robust and interpretable forecasts, though it also faces challenges with complex nonlinear effects unless hybridized (Taylor and Letham, 2018). Overall, statistical models are strong and transparent but less accurate in modern, complex load forecasting scenarios (Weron, 2006).

2.2. Machine learning approach

Machine learning (ML) offers a flexible progression beyond classical statistical models for electricity load forecasting by capturing nonlinear relations between demand and exogenous drivers such as weather, calendar effects, and price signals (Jain and Gupta, 2024; Li et al., 2024). Among shallow learners, Support Vector Regression and Random Forests are popular for their robustness on modest datasets and for modeling interaction effects (Fan et al., 2021). Gradient-boosted trees (XGBoost, LightGBM) further improve accuracy and scalability in high-dimensional settings through regularized boosting (Shehadeh et al., 2021). Deep learning (DL) extends ML by learning hierarchical temporal representations that cover both short- and long-range dependencies (Chen et al., 2023): RNN variants (LSTM/GRU) perform well for short- to mid-term horizons and volatile demand (ArunKumar et al., 2022; Abumohsen et al., 2023). At the same time, CNNs extract local temporal/spatial patterns and often complement recurrent models (Koprinska et al., 2018). More recently, attention-based architectures, such as Transformers and the Temporal Fusion Transformer, have achieved state-of-the-art accuracy by modeling long-range dependencies without recurrence and by integrating variable selection and interpretable multi-head attention (Lim et al., 2021; Wu et al., 2022). Overall, modern ML/DL approaches typically outperform traditional baselines in multi-horizon, weather-sensitive settings, while also enabling probabilistic forecasts (e.g., via quantile losses) for risk-aware decision-making (Lim et al., 2021; Wu et al., 2022).

2.3. Hybrid approach

Hybrid models that combine complementary architectures have shown strong potential in electricity load forecasting. CNN–LSTM frameworks are widely used (Zha et al., 2022), with CNN layers extracting short-term spatial–temporal features and LSTM layers capturing sequential dependencies, resulting in more robust predictions under fluctuating demand conditions (Liu et al., 2021). Extending this design, CNN–BiLSTM hybrids improve accuracy by modeling bidirectional temporal dynamics, which enhances sensitivity to both past and future contextual patterns within the input sequence (Zhang and Wang, 2023). Transformer-based hybrids have gained prominence for multi-horizon forecasting, as their attention mechanisms efficiently capture long-range dependencies while maintaining scalability (Wen et al., 2022), particularly when integrated with decomposition methods such as Prophet or VMD to separate trend and residual signals (Song et al., 2022a). In parallel, boosting-based hybrids, including XGBoost and LightGBM combined with neural networks, provide competitive performance by effectively handling high-dimensional inputs, mitigating overfitting, and delivering improved generalization across diverse operating conditions (Hamed et al., 2023). Collectively, these hybrids advance forecasting accuracy beyond standalone models, though their computational complexity and data requirements remain active challenges for real-time applications.

3. Dataset overview

In this study, we use Jeju Island's hourly electricity load data (South Korea) together with meteorological observations from four weather stations: Jeju-si, Gosan, Seongsan, and Seogwipo. The dataset spans 2012–2020 and reports the total load for the Jeju region. Fig. 2 shows average monthly load patterns over the nine-year period, capturing long-term trends and seasonal variability. Each station provides meteorological variables including air temperature (TA), dew point (TD), humidity (HM), wind speed (WS), wind direction (WD), atmospheric pressure (PA), discomfort index (DI), sensible temperature (ST), and solar radiation (SI).

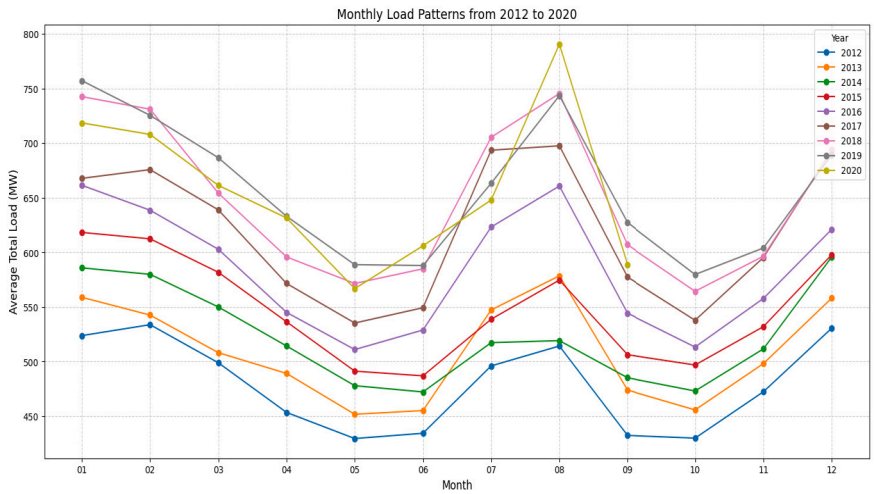


Fig. 2. Monthly electricity load patterns from 2012 to 2020. (Each line represents a specific year, highlighting consistent peaks in summer (August) and winter (January–February), associated with cooling and heating demands, respectively.)

Table 1
Summary of recent approaches in electricity load forecasting, highlighting methods, contributions, and limitations.

Paper	Methods used	Contribution	Limitation
Tarmanini et al. (2023)	ARIMA, ANN	Compared ARIMA and ANN on household load; ANN handled nonlinear demand more accurately.	ARIMA weak for nonlinear data; ANN overlooked peak loads.
Zhang and Wang (2023)	MA–CNN–BiLSTM–DE	Developed deep ensemble with multi-head attention; improved accuracy and uncertainty quantification.	High model complexity; computationally demanding.
Shehadeh et al. (2021)	XGBoost, LightGBM	Boosting methods achieved strong generalization and high forecasting accuracy.	Sensitive to hyperparameters; potential overfitting.
Abumohsen et al. (2023)	LSTM, GRU, RNN	Evaluated RNN family for short-term load forecasting; GRU under Adam optimizer performed best.	Requires large datasets; limited interpretability.
Lim et al. (2021)	TFT	Introduced interpretable multi-horizon forecasting with variable selection and attention mechanisms.	High computational cost.
Fan et al. (2024)	DTformer (Transformer + Dual Attention + TWA)	Optimized Transformer with dual variable attention and time-aware attention; achieved SOTA multi-energy forecasting.	Still resource-intensive; weaker in short-term accuracy.
Bashir et al. (2022)	Prophet–LSTM + BPNN	Hybrid of Prophet, LSTM, and BPNN; reduced RMSE, MAE, and MAPE significantly.	Complex architecture; slower training.
Wan et al. (2023)	CNN–LSTM	Combined spatial–temporal extraction with sequence modeling; improved short-term accuracy.	Limited interpretability.
Gong et al. (2025)	TCN–LSTM + LightGBM	Ensemble combining DL and boosting; robust across regions and load types.	High memory and computation needs; weaker on irregular holiday data.

Table 2
Descriptive statistics of the Jeju hourly energy consumption dataset.

Feature	Min	Max	Mean	Std	CV	Unit
Total_Load	0.00	1018.00	578.63	108.72	0.188	MW
Total_TA	−5.75	33.50	16.45	7.71	0.469	°C
Total_TD	−11.50	28.25	11.45	9.29	0.811	°C
Total_WS	0.00	23.00	3.69	1.76	0.477	m/s
Total_WD_DEG	0.00	360.00	198.96	78.04	0.392	°
Total_PA	969.50	1032.00	1011.12	7.67	0.008	hPa
Total_DI	25.50	86.25	61.55	11.96	0.194	–
Total_ST	−13.50	33.50	15.76	8.72	0.553	°C
Total_SI	0.00	4.00	0.96	1.00	1.040	MJ/m^2

3.1. Data analysis

We analyze the relationship between weather conditions and electricity consumption on Jeju Island using hourly energy data from 2012 to mid-2020. The dataset includes total energy consumption (Total Load) and various meteorological features from four climate stations: Jeju-si, Gosan, Seongsan, and Seogwipo.

The descriptive statistics, as shown in Table 2, outline the primary attributes used in this study. These features are crucial for predicting Total Load, which is the target variable. For each feature, the following statistical metrics were calculated: Coefficient of Variation (CV), Mean, Standard Deviation (Std), Minimum (Min), and Maximum (Max). The CV shows each feature's relative variability; low variability is indicated by values less than 1. The Total Load variable indicates considerable energy consumption fluctuation, with a mean of 578.63 MW and a

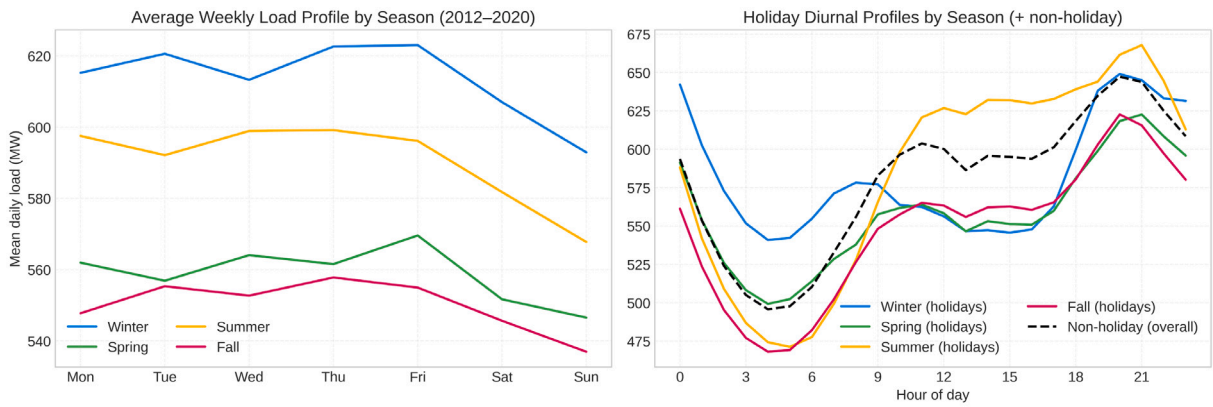


Fig. 3. Weekly and holiday seasonality of Jeju load (2012–2020). The left panel shows mean daily load by day of week for each season: weekdays exceed weekends, winter is highest overall, and summer shows a mid-week hump. The right panel plots holiday diurnals against a non-holiday baseline (dashed), revealing depressed daytime demand but an evening rebound on holidays, strongest in winter and summer.

Table 3
Calendar variables and South Korean public holidays used in the analysis (2012–2020).

Calendar variable	Encoding	Rationale/expected load effect	KR public-holiday examples used in the flag[a]
`hour` (0–23)	integer; also sine/cosine pair	Captures diurnal cycle (morning ramp, evening peak; night valley).	–
`day_of_week` (0–6)	one-hot	Weekday vs weekend profile; business activity drops on Sat/Sun.	–
`is_weekend`	boolean	Weekend flattening; lower daytime commercial load.	–
`month` (1–12)	one-hot	Coarse seasonal shift in baseline due to heating/cooling needs.	–
`season` (Winter, Spring, Summer, Fall)	one-hot	Heating (Winter: Dec–Feb) and cooling (Summer: Jun–Aug) regimes.	–
`is_holiday`	boolean (from `holidays.KR`)	Public holidays behave like Sundays; reduced industrial/commercial demand; sometimes evening leisure peaks.	New Year's Day (Jan 1), Independence Movement Day (Mar 1), Children's Day (May 5), Buddha's Birthday (lunar), Memorial Day (Jun 6), Liberation Day (Aug 15), National Foundation Day (Oct 3), Hangul Day (Oct 9), Christmas (Dec 25), **Seollal** (LNY, 3-day, lunar), **Chuseok** (Mid-Autumn, 3-day, lunar), substitute holidays.
`is_pre_holiday`	boolean (day before any holiday)	Pre-holiday shopping/travel uplift; evening peak often stronger.	As above (especially Seollal, Chuseok).
`is_post_holiday`	boolean (day after any holiday)	Rebound/return-to-work effects; morning peak recovery.	As above.
`is_year_end`	boolean (Dec 31–Jan 1 window)	Year-end closures and leisure effects; atypical late-evening usage.	New Year's period.

[a] Holiday dates follow the official South Korea calendar, including substitute-holiday rules. Lunar holidays (Seollal and Chuseok) shift by year.

standard deviation of 108.72 MW. The Total-TA (Temperature) and Total-TD (Dew Point Temperature) show significant seasonal fluctuations, with Total-TA ranging from −5.75 °C to 33.50 °C. Total-WS (Wind Speed) ranges from 0.00 to 23.00 m/s, with a mean of 3.69 m/s.

Features such as Total-DI (Discomfort Index) and Total-ST (Sensible Temperature) reflect more specialized meteorological conditions. Total-DI ranges from 25.50 to 86.55, indicating variations in discomfort related to temperature and humidity, while Total-ST ranges from −13.50 °C to 35.50 °C, reflecting seasonal temperature shifts. For For Total-PA (Pressure), the mean is 1011.12 hPa with a low CV, indicating relatively stable pressure across the dataset. Beyond weather, short to mid-term electricity demand in Jeju is strongly calendar-driven. We therefore engineer and analyze calendar variables alongside an official South Korea holiday flag (from `holidays.KR`). Table 3 summarizes the variables and their expected effects. Concretely, load exhibits a pronounced *hour-of-day* cycle (morning/evening peaks), a *day-of-week* effect (weekend flattening), clear *monthly/seasonal* shifts (winter heating and summer cooling), and systematic *holiday* behavior (most public holidays resemble Sundays, with softer business-hour demand). We also include *pre-holiday* and *post-holiday* flags to capture shopping/travel surges before, and rebound effects after, multi-day holidays such as Seollal and Chuseok.

Fig. 3 shows that a clear calendar structure governs Jeju's load. On the left, the average weekly profile reveals a high winter baseline, next-highest summer, and lower spring/fall, with a consistent weekday plateau (Mon–Fri) and a pronounced weekend drop (Sat–Sun) strong day-of-week effects that persist across seasons. On the right, holiday diurnals differ systematically from the non-holiday benchmark (dashed): demand is markedly lower during business hours (≈ 09–≈ 16) and shifts toward stronger evening peaks (≈ 19–≈ 22), with the largest holiday suppression in winter and summer and milder effects in spring/fall. Together, the panels confirm that season, day-of-week, and holidays reshape both the level and the intraday shape of load, underscoring the value of calendar features (including holiday flags) in forecasting. Fig. 4 presents density plots of weather features and total energy consumption across the four stations. The plot reveals that temperature (TA) exhibits the broadest distribution at Jeju-si, while Seogwipo shows a narrower range. Dew point (TD) and humidity (HM) are notably lower at Gosan and Seogwipo. Seogwipo also experiences the highest wind speed (WS), while wind direction (WD-DEG) displays greater variability at this station. Atmospheric pressure (PA) remains relatively consistent across stations, with slightly higher values at Gosan and Seongsan. Additionally, the discomfort index (DI) is higher at Jeju-si and Seogwipo, whereas the sensible temperature (ST) is highest in

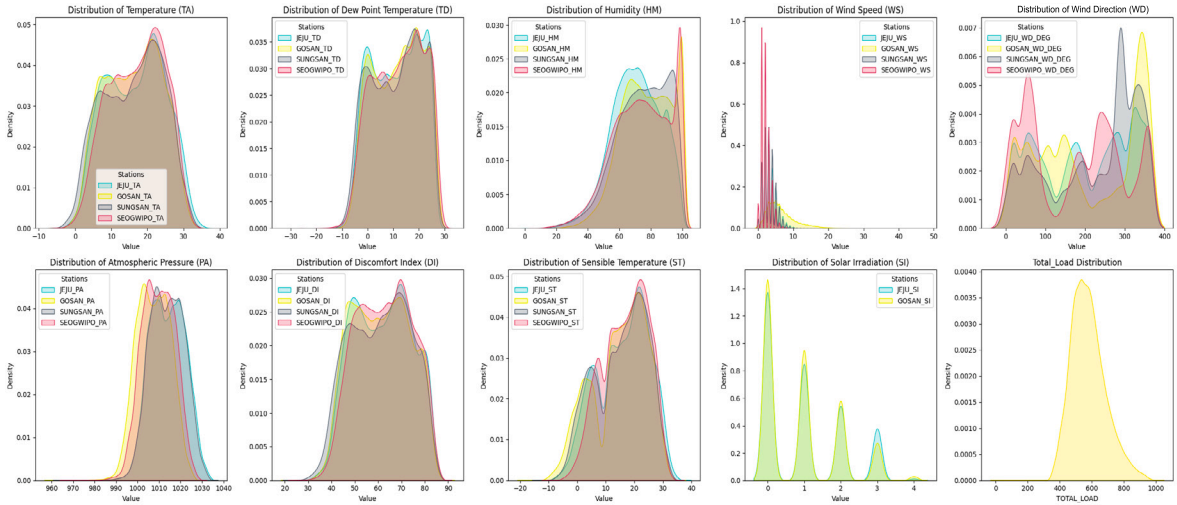


Fig. 4. Distribution of meteorological features and total load data across different stations. The distribution plots illustrate the variations in each feature's values across the stations, offering insight into the seasonal trends and differences in weather patterns that impact energy consumption.

Gosan. Solar irradiation (SI) is most pronounced at Jeju-si. Finally, Total Load follows a bell curve distribution, with values concentrated around 600 MW, slightly higher at Jeju-si, reflecting the island's overall energy consumption patterns.

The analysis shows that electricity use on Jeju Island varies with seasonal and weather factors, with clear station-level differences in temperature, wind, and solar irradiation. While these findings are exploratory and not statistically tested, they point to the need for modeling daily, weekly, annual, and holiday seasonalities. In the proposed framework, these factors are formally incorporated, and feature importance analysis provides quantitative evidence of their impact.

3.2. Data processing

The data preprocessing begins with initial cleaning steps. We first drop only columns that are entirely missing (`SEONGSAN-SI` and `SEOGWIPO-SI`), which carries no statistical cost. To preserve the weekly structure (7 × 24 hours), we do not remove any time steps. Instead, we build a complete hourly index from the first to the last timestamp and reindex the data to this grid, parsing the stamp `D_BASE_DT` accordingly.

```
pd.to_datetime(D_{BASE_DT}, format='%d-%m-%y %H:%M')
```

The datetime column then sorts the data, and the index is reset to maintain proper temporal order. For numerical features $X \in \mathbb{R}^{n \times k}$, the total missing values are only 3.10%, so they are imputed using mean imputation per feature (Du et al., 2020). For each column x_j, the imputed value is the average over its observed entries:

$$\hat{x}_j = \frac{1}{n_j} \sum_{i \in S_j} x_{ij}, \qquad S_j = \{\, i \mid x_{ij} \neq \mathrm{NaN} \,\}, \qquad n_j = |S_j|. \tag{1}$$

Here, n_j is the number of non-missing values in column j after reindexing to the hourly grid. Because all hours are retained and only the values at missing hours are imputed (no records deleted), the temporal structure, daily/weekly seasonality, and calendar alignment remains intact.

To extract features from the dataset, we employed the Prophet model (Ma et al., 2023), which breaks down the target time series $y(t)$ into four additive components: $g(t)$ for long-term trend, $s(t)$ for periodic seasonality, $h(t)$ for holiday effects, and ϵ_t for random noise. The model is expressed as:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{2}$$

Here, $g(t)$ models non-periodic trends, $s(t)$ captures repeated seasonal cycles, $h(t)$ accounts for special events, and ϵ_t represents the residual error. The extracted trend $\hat{g}(t)$ and seasonality $\hat{s}(t)$ are used as additional features.

From the datetime object, we derived time-based features such as hour (0–23), day of the month (1–31), month (1–12), day of the week (0–6; where 0 = Monday), and a weekend flag (1 for Saturday or Sunday, 0 otherwise). Additionally, weather-related features $W = \{w_1, w_2, \ldots, w_k\}$ are retained, and lag features are created to incorporate past information, including:

$$\mathrm{lag}_1 = y_{t-1} \tag{3}$$

$$\mathrm{lag}_{24} = y_{t-24} \tag{4}$$

Lag1 captures immediate hourly autocorrelation, while Lag24 captures daily cyclicity. To maintain parsimony and reduce collinearity, we restricted the lag set in this work. A continuous time index is constructed as:

$$\mathrm{time_idx}_t = \left\lfloor \frac{t - t_0}{3600\,\mathrm{seconds}} \right\rfloor \tag{5}$$

where t_0 is the timestamp of the dataset's starting point. Outlier removal is an essential step to confirm the dataset's quality before the training process. In this study, outliers in the target variable `TOTAL_LOAD` are identified and removed using the Interquartile Range (IQR) method. The IQR is calculated as:

$$IQR = Q3 - Q1 \tag{6}$$

where: Q1 is the first quartile (25th percentile) and Q3 is the third quartile (75th percentile). In order to identify extreme outliers, the lower and upper boundaries are defined as:

$$\text{Lower Bound} = Q1 - 3.5 \times IQR \tag{7}$$

$$\text{Upper Bound} = Q3 + 3 \times IQR \tag{8}$$

Any data points $t_{\mathrm{TOTAL_LOAD}}$ falling outside these bounds are considered extreme outliers and are removed. We chose removal rather than imputation (e.g., neighbor averaging) to avoid injecting artificial values that could bias the short-term autocorrelation structure. Only $\approx 0.7\%$ of hourly load records were removed as outliers; this minimal removal preserves temporal structure, while Prophet's decomposition and calendar covariates mitigate the limitations of IQR on non-stationary load data. The resulting cleaned dataset is defined as,

$$D_{\mathrm{processed}} = \{t \mid \text{Lower Bound} \leq t_{\mathrm{TOTAL_LOAD}} \leq \text{Upper Bound}\} \tag{9}$$

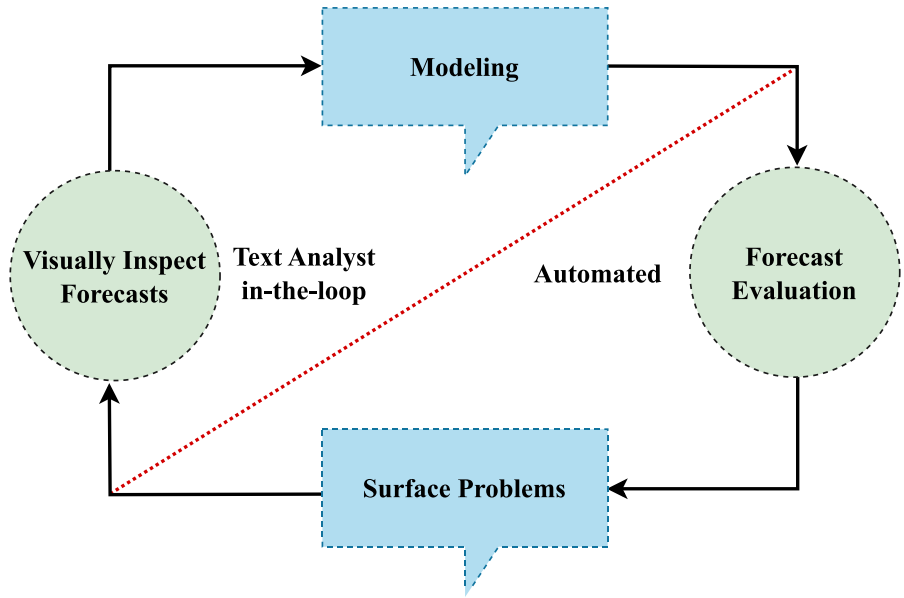


Fig. 5. Workflow of the Prophet model. The diagram illustrates the iterative process of Prophet's forecasting methodology, including modeling, forecast evaluation, surfacing problems, and visual inspection, with a human-in-the-loop (TextAnalyst) aiding in refinement.

After outlier removal, a training set and a testing set of the dataset are separated, based on a temporal cutoff:

$$D_{\text{train}} = \{t \mid t < 2019\text{-}01\text{-}01\} \tag{10}$$

$$D_{\text{test}} = \{t \mid t \geq 2019\text{-}01\text{-}01\} \tag{11}$$

The `time_idx` in the test set is adjusted to ensure continuity with the training set by:

$$\text{time_idx}_{\text{test}} = \max(\text{time_idx}_{\text{train}}) + 1 + i, \quad \forall i \in [0, |D_{\text{test}}|) \tag{12}$$

where $|D_{\text{test}}|$ denotes the size of the test set. Using this time index, we then split the data into training and testing sets based on a specific cutoff date, maintaining the temporal order. Although Prophet models global trend explicitly, `time_idx` serves a different role: it provides a monotonic positional encoding for the TFT, preserving temporal order across train–test splits and rolling-origin evaluation. This ensures continuity for lagged features and covariates, allowing the neural component to capture local sequential dynamics beyond the global trend.

4. Methodology

4.1. Prophet model

The time series forecasting model called Prophet was created by Meta's (Facebook) Data Science team (Taylor and Letham, 2018). It is intended to manage data from time series with several seasonalities, missing values, outliers, and abrupt trend changes (Chaturvedi et al., 2022). Instead of relying on deep learning architectures, Prophet leverages classical statistical modeling with a Bayesian framework (Wang et al., 2024a). Fig. 5 illustrates Prophet's method of modeling trend, seasonality, and holiday effects for time-series forecasting.

Prophet breaks down a time series into the primary parts (Wang et al., 2024b) listed below: **trend** ($g(t)$), **seasonality** ($s(t)$), and **holidays or special events** ($h(t)$), with an added **error term** ϵ_t to account for noise:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{13}$$

Trend Component: The trend component $g(t)$ models non-periodic, long-term changes in the data. Prophet supports both linear and logistic growth models (Toharudin et al., 2023). One commonly used form is the logistic growth function:

$$g(t) = \frac{C}{1 + \exp(-k(t - m))} \tag{14}$$

Here, C is the carrying capacity, k is the growth rate, and m is the offset parameter that shifts the curve along the time axis. This kind of setup enables the model to capture saturating growth behavior that is widely observed in real-time series. For the inclusion of abrupt changes in the trend, Prophet utilizes a piecewise logistic growth model with changepoints (Zhao et al., 2022):

$$g(t) = \frac{C(t)}{1 + \exp\left(-\left(k + \mathbf{a}(t)^\top \boldsymbol{\delta}\right)\left(t - \left(m + \mathbf{a}(t)^\top \boldsymbol{\gamma}\right)\right)\right)} \tag{15}$$

In this equation, $\mathbf{a}(t)$ is a binary indicator vector that activates when t is after a changepoint, $\boldsymbol{\delta}$ adjusts the growth rate, and $\boldsymbol{\gamma}$ adjusts the offset at those changepoints. The values of $\boldsymbol{\delta}$ are regularized using a Laplace prior:

$$\delta_j \sim \text{Laplace}(0, \tau) \tag{16}$$

This prior promotes sparsity, so only significant changes in the trend are incorporated, allowing for robust modeling of structural shifts without overfitting.

Seasonality Component: The seasonality component $s(t)$ captures recurrent patterns in the data (Naima et al., 2023), including cycles that occur every day, every week, or every year.

$$s(t) = \sum_{n=1}^{N} \left[a_n \cos\left(\frac{2\pi nt}{P}\right) + b_n \sin\left(\frac{2\pi nt}{P}\right)\right] \tag{17}$$

In this equation, P is the known time frame of the seasonal cycle (365.25 for yearly seasonality), N is the Fourier order (which determines the number of sine and cosine terms used), and a_n, b_n are learned coefficients. This flexible formulation allows Prophet to effectively model complex seasonal behaviors.

Holiday Component: The holiday element $h(t)$ is responsible for the influence of holidays or unique occasions that can cause temporary deviations in the time series. These effects are modeled as:

$$h(t) = \sum_{j} \sum_{h=0}^{23} \kappa_{j,h} \, \mathbf{1}\{t \in \text{holiday } j\} \, \mathbf{1}\{\text{hour}(t) = h\}. \tag{18}$$

Here, $H_j(t)$ is a day-level holiday indicator equal to 1 if the calendar date of t is holiday j (including substitute days) and 0 otherwise;

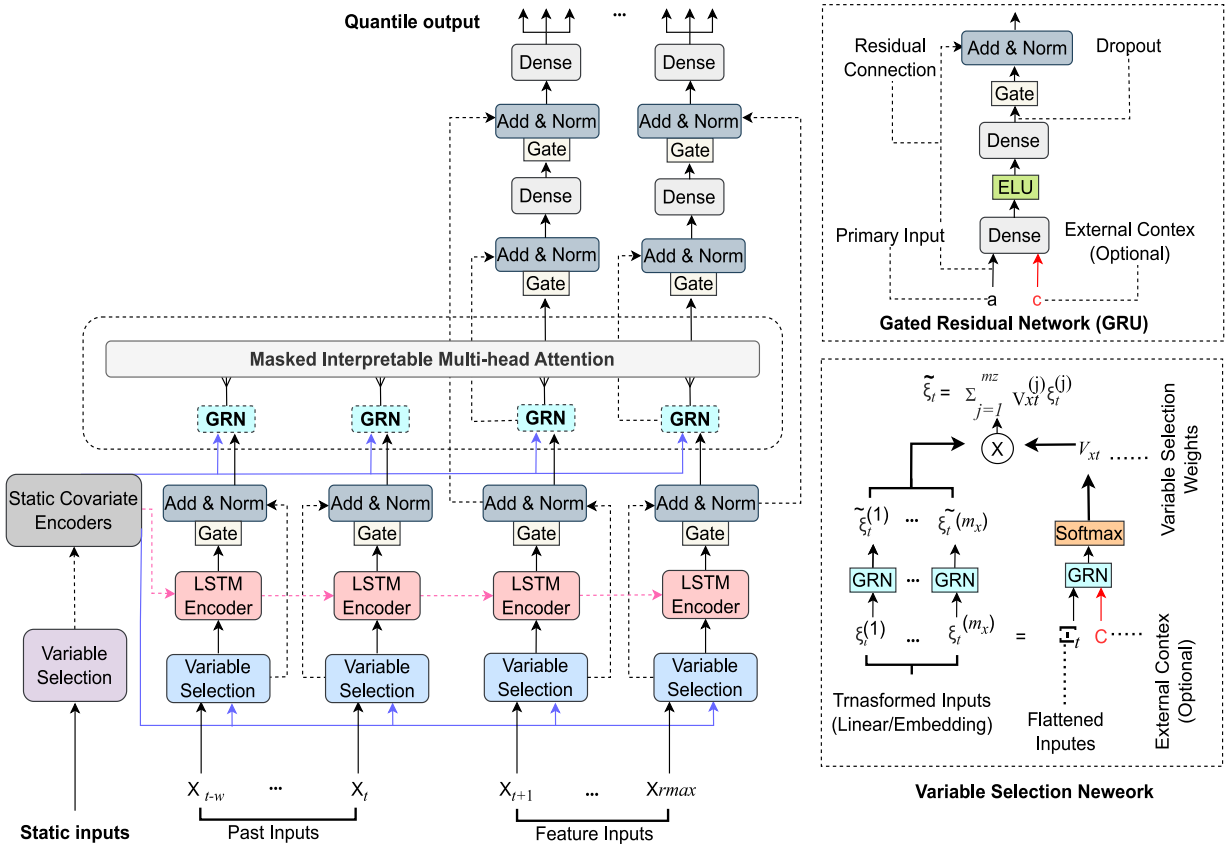


Fig. 6. Temporal Fusion Transformer (TFT) architecture for multi-horizon forecasting. The figure shows how static, past, and future inputs are processed through variable selection. It also represents how LSTM encoders and multi-head attention are used to generate probabilistic forecasts, combining accuracy with interpretability.

$\mathbf{1}\{\text{hour}(t) = h\}$ selects hour h of the day; and $\kappa_{j,h}$ is the estimated hour-specific effect size for holiday j. This specification allows holiday impacts to vary by time of day, capturing intraday spikes or dips (Rafferty, 2023).

Error Term: The error term ϵ_t captures random noise and differences in the data that are not described by other components. It is expected to be frequently distributed and demonstrate that the model accounts for uncertainty and unpredictable fluctuations in real-world observations. The error term is crucial for producing probabilistic forecasts, where the model provides a distribution of possible future values instead of a single point estimate.

4.2. Temporal fusion transformer (TFT)

A state-of-the-art deep learning framework for multi-horizon, interpretable time series forecasting is called the Temporal Fusion Transformer (Nazir et al., 2023). Developed to manage diverse inputs and intricate temporal patterns (Wu et al., 2022), the Temporal Fusion Transformer (TFT) combines the benefits of attention mechanisms with recurrent neural networks (Dai et al., 2025). As illustrated in Fig. 6, TFT integrates five key components: Static Covariate Encoders, LSTM-based Temporal Encoder–Decoder, Variable Selection Networks, Gated Residual Networks (GRNs), and Interpretable Multi-Head Attention, together with Quantile Output Layers to balance interpretability, input adaptability, and forecasting accuracy (Lim et al., 2021; Li et al., 2023).

Gated Residual Networks (GRNs) allow selective transformation of inputs with optional contextual information. The GRN first computes an intermediate hidden state:

$$h_2 = \text{ELU}(W_{2,u}a + W_{3,u}c + b_{2,u}) \tag{19}$$

where a is the input vector, c is an optional context, and ELU is the exponential linear unit. This is linearly projected:

$$h_1 = W_{1,u}h_2 + b_{1,u} \tag{20}$$

Then, a Gated Linear Unit (GLU) controls the information flow:

$$\text{GLU}(h_1) = \sigma(W_{4,u}h_1 + b_{4,u}) \odot (W_{5,u}h_1 + b_{5,u}) \tag{21}$$

Here, σ is the sigmoid activation, and $\odot$ is element-wise multiplication. Finally, the GRN output is:

$$\text{GRN}(a, c) = \text{LayerNorm}(a + \text{GLU}(h_1)) \tag{22}$$

This ensures stable training via residual connection and layer normalization, while allowing non-linear gating.

Variable Selection Networks dynamically identify which input variables are most important (Zhou et al., 2024). For a time step t, the flattened input I_t and static context c_s are passed into a GRN:

$$V_{x_t} = \text{Softmax}(\text{GRN}_v([I_t; c_s])) \tag{23}$$

Each individual variable $x_t^{(j)}$ is transformed:

$$\tilde{x}_t^{(j)} = \text{GRN}(x_t^{(j)}) \tag{24}$$

and all transformed variables are weighted by their learned importance and aggregated:

$$\tilde{x}_t = \sum_{j=1}^{m_x} v_t^{(j)} \cdot \tilde{x}_t^{(j)} \tag{25}$$

This enables the model to focus on relevant features and ignore noise.

Interpretable Multi-Head Attention captures long-term temporal dependencies between time steps. First, attention weights are computed

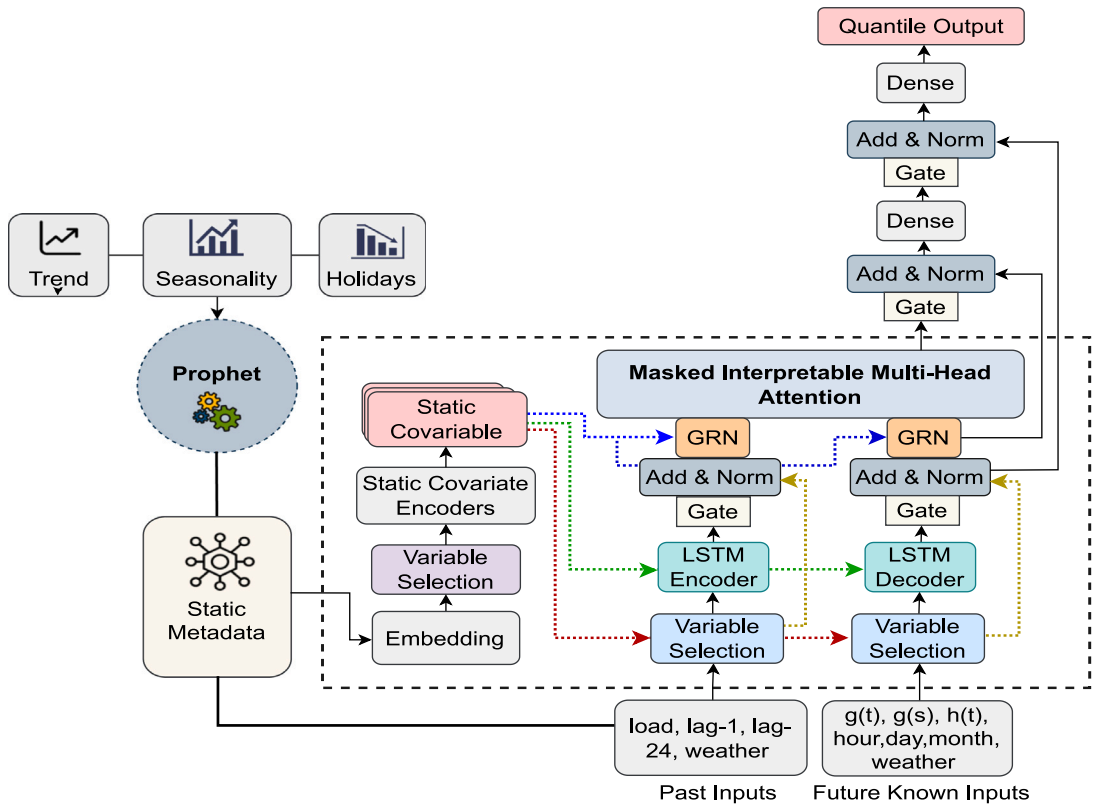


Fig. 7. Proposed architecture of the hybrid Prophet–TFT model for multi-step probabilistic load forecasting.

using scaled dot-product attention:

$$A(Q,K) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{\text{attn}}}}\right) \tag{26}$$

Each attention head h uses transformed queries, keys, and shared values:

$$H_h = \text{Attention}(QW_Q^h, KW_K^h, VW_V) \tag{27}$$

The attention outputs are averaged:

$$\tilde{H} = \frac{1}{m_H}\sum_{h=1}^{m_H} A(QW_Q^h, KW_K^h)VW_V \tag{28}$$

and then projected:

$$\text{Output} = \tilde{H}W_H \tag{29}$$

This modification ensures interpretability, as all attention heads use the same value projection VW_V, allowing consistent analysis of temporal dependencies.

Quantile Output Layer enables uncertainty modeling by predicting quantile forecasts (e 10th, 50th, 90th percentiles). The model is trained using a quantile loss function:

$$\text{QL}(y,\hat{y};q) = q(y-\hat{y})^+ + (1-q)(\hat{y}-y)^+ \tag{30}$$

where $(x)^+ = \max(0,x)$. The total loss aggregates across time steps, samples, and quantiles:

$$L(U,W) = \frac{1}{MT_{\max}}\sum_{y_t\in U}\sum_{q\in Q}\sum_{T=1}^{T_{\max}} \text{QL}(y_t,\hat{y}_{t,T(q)};q) \tag{31}$$

TFT not only models complex temporal relationships and non-linear dynamics, but also provides interpretability via variable selection and attention analysis, making it highly effective for real-world applications in energy forecasting, finance, and healthcare. TFT's modularity makes it capable of multi-horizon forecasting (Fan et al., 2024), where several steps in the future are predicted in a single forward pass. Merging static features, temporal patterns, attention, and real-time dynamic weather data ensures that the model produces precise, flexible, and explainable forecasts (Peng et al., 2023). It is especially suited for energy demand forecasting, wherein utilities and grid operators require simultaneous short-term and mid-term predictions to maximize energy production, distribution, and pricing strategies.

4.3. Proposed method (Prophet-TFT model)

The proposed hybrid Prophet–Temporal Fusion Transformer (TFT) model effectively combines the interpretability of classical time series decomposition with the powerful temporal modeling capabilities of deep learning. Fig. 7 shows the overall architecture of the proposed model, demonstrating how Prophet's decomposed components are integrated into the Temporal Fusion Transformer framework for enhanced multi-horizon forecasting.

The modeling process begins with the Prophet model, which is applied to the original time series $y(t) \in \mathbb{R}$ for time steps $t = 1, 2, \ldots, T$. Prophet decomposes (Stefenon et al., 2023) $y(t)$ as:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t, \tag{32}$$

where $g(t)$ is the non-linear growth trend component, $s(t)$ is the seasonal component modeled as:

$$s(t) = \sum_{k=1}^{K}\left[a_k\cos\left(\frac{2\pi kt}{P}\right) + b_k\sin\left(\frac{2\pi kt}{P}\right)\right],$$

with P as the period, $h(t)$ encodes holidays via indicator functions, and $\varepsilon_t \sim \mathcal{N}(0,\sigma^2)$ is Gaussian noise (Stefenon et al., 2023). The components $\hat{g}(t), \hat{s}(t), \hat{h}(t)$ are not used for direct prediction, but as auxiliary inputs into the TFT. It processes three types of inputs: static metadata, past observed inputs, and future known inputs (Zhou et al., 2024). Lag features (lag1 and lag24) are recomputed at each rolling origin using the most recent historical values, ensuring that forecasts for 2019–2020

rely only on observed data and not on recursive imputation. Static metadata $X_{\text{static}} \in \mathbb{R}^{N\times d_s}$ are embedded as:

$$e_{\text{static}} = \text{Embedding}(X_{\text{static}}) \in \mathbb{R}^{N\times d_e}, \tag{33}$$

serving as global conditioning signals for all subsequent layers. Past inputs $X_{\text{past}} \in \mathbb{R}^{N\times T_p\times d_p}$ include lag features, historical targets, and weather. They are processed by the Variable Selection Network (VSN):

$$X'_{\text{past}} = \text{VSN}(X_{\text{past}}), \tag{34}$$

Similarly, future known inputs $X_{\text{future}} \in \mathbb{R}^{N\times T_f\times d_f}$, which include Prophet's decomposed outputs $\hat{g}(t), \hat{s}(t), \hat{h}(t)$, calendar covariates (such as hour, day, month), and forecasted weather variables, are passed through a separate VSN:

$$X'_{\text{future}} = \text{VSN}(X_{\text{future}}). \tag{35}$$

The selected outputs of X'_{past} and X'_{future} are then directed into the LSTM encoder and an LSTM decoder branch of the TFT. The LSTM encoder captures temporal dependencies from historical data, while the decoder focuses on modeling future known inputs (Koya and Roy, 2024), facilitating accurate multi-step forecasting.

$$h_{\text{enc}} = \text{LSTMEncoder}(X'_{\text{past}}) \in \mathbb{R}^{N\times T_p\times d_h}, \tag{36}$$

$$h_{\text{dec}} = \text{LSTMDecoder}(X'_{\text{future}}) \in \mathbb{R}^{N\times T_f\times d_h}, \tag{37}$$

where d_h is the hidden state size. The encoder and decoder outputs are fused using a masked multi-head attention mechanism:

$$h_{\text{attn}} = \text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W^O, \tag{38}$$

where:

$$\text{head}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}} + M\right)V_i, \tag{39}$$

and Q, K, V are projections from h_{dec} and h_{enc}, and M is a causal mask matrix to preserve autoregressive structure.

The attention outputs h_{attn} are passed through Gated Residual Networks:

$$h_{\text{GRN}} = \text{LayerNorm}(x + \sigma(W_1 x + b_1) \cdot \text{ELU}(W_2 x + b_2)), \tag{40}$$

which improves nonlinearity and stabilizes optimization. Finally, the refined representation h_{GRN} is processed through position-wise feed-forward and dense layers to yield quantile forecasts:

$$\hat{y}^{(\alpha)}_{t+\tau} = \text{Dense}(\text{FeedForward}(h_{\text{GRN}})), \quad \alpha \in \{0.1, 0.5, 0.9\}, \tag{41}$$

where τ is the future time step and α denotes the quantile level. By accurately incorporating Prophet's seasonal decomposition and processing future covariates through a dedicated decoder path, the proposed model provides uncertainty-aware multi-horizon forecasts and yields lower errors with well-calibrated intervals, ideal for predicting electricity loads in dynamic seasonal contexts.

To curb overfitting, especially on rare holiday weather interactions, we adopt a rolling-origin, expanding-window evaluation: at each origin, the model is trained on the past and produces 1–24h forecasts for the next week, with origins stepped every four weeks; major holiday windows (Seollal/Chuseok) are held out entirely. To prevent leakage, Prophet is re-fit within each fold using only training timestamps, and its components $\hat{g}(t), \hat{s}(t), \hat{h}(t)$ for validation/test are generated from that fold; decoder "known-future" inputs are limited to calendar/holiday flags and other ex-ante covariates. Model capacity is regularized via dropout (0.2–0.3), L2 (10^{-4}), layer normalization and gradient clipping, with early stopping on validation CRPS and adaptive learning-rate reduction. Expressivity is bounded (1–2 LSTM layers, d_h = 64–128, 4 heads), and TFT's variable-selection gates suppress weak covariates; results are averaged over three seeds. This time-aware protocol and explicit regularization yield robust generalization without sacrificing temporal structure.

Table 4
Compute environment used for all model training and evaluation, detailing GPU, CPU, operating system, and memory configuration.

Hardware and software configuration for model training	
GPU	NVIDIA GeForce RTX 4060 (8 GB Dedicated)
CPU	13th Gen Intel® Core™ i7-13700 @ 2.10 GHz
Operating System	Windows 11 Home
RAM	16.0 GB DDR5 @ 5600 MHz

4.4. Model training and evaluation

To ensure a fair comparison, all forecasting models were trained and evaluated under the same experimental setup. The dataset spans 2012–2020, with 2012–2018 used for training and 2019–2020 reserved for testing. This temporal division reflects a realistic scenario in which models are fitted only on historical data and then applied to unseen future demand. Fig. 8 illustrates the split: the blue segment corresponds to the training period (2012–2018), while the red segment denotes the test period (2019–2020).

We evaluate the proposed Prophet–TFT hybrid against existing models, including FFNN (Stienecker and Hagemeier, 0000), CNN–LSTM (Coutinho et al., 2025), XGBoost–LSTM (Xue et al., 2022), GAMs (Gaillard et al., 2016), EMD–LSTM (Mounir et al., 2023), and Prophet–LSTM (Bashir et al., 2022). Table 4 summarizes the computing environment, while Table 5 details the main parameter settings. For Prophet, this includes seasonality, changepoint, and feature-engineering options; for TFT, it covers network architecture, training specifications, and quantile loss. This unified setup ensures that observed performance differences are attributable to model design rather than inconsistencies in training or evaluation.

We assess performance with point and probabilistic metrics. Point metrics are computed from the median (P50) forecast: MAE (mean absolute error):

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{42}$$

MAE ignores error direction but provides a simple measure of average deviation.

RMSE (root mean squared error):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{43}$$

RMSE is highly sensitive to outliers yet highlights large errors effectively.

MAPE (mean absolute percentage error):

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{44}$$

MAPE is undefined when $y_i = 0$ but allows intuitive percentage-based comparisons. Probabilistic quality is evaluated from the full predictive distribution using CRPS, Pinball loss, coverage (P50, P90 nominal rates), and sharpness (average interval width). CRPS (Continuous Ranked Probability Score) generalizes MAE to probabilistic forecasts by scoring the entire predictive distribution rather than a single point estimate, rewarding both sharpness and calibration. For a quantile level α, the pinball loss is,

$$\mathcal{L}_\alpha(y, \hat{y}^\alpha) = \begin{cases} \alpha\,(y - \hat{y}^\alpha), & y \geq \hat{y}^\alpha \\ (1-\alpha)\,(\hat{y}^\alpha - y), & y < \hat{y}^\alpha \end{cases} \tag{45}$$

Pinball loss measures quantile forecasting accuracy, penalizing under and over-predictions asymmetrically depending on α. All metrics are computed on the 2019–2020 test period; monthly/seasonal tables aggregate repeated short-term (1–24 h) forecasts made at rolling origins.

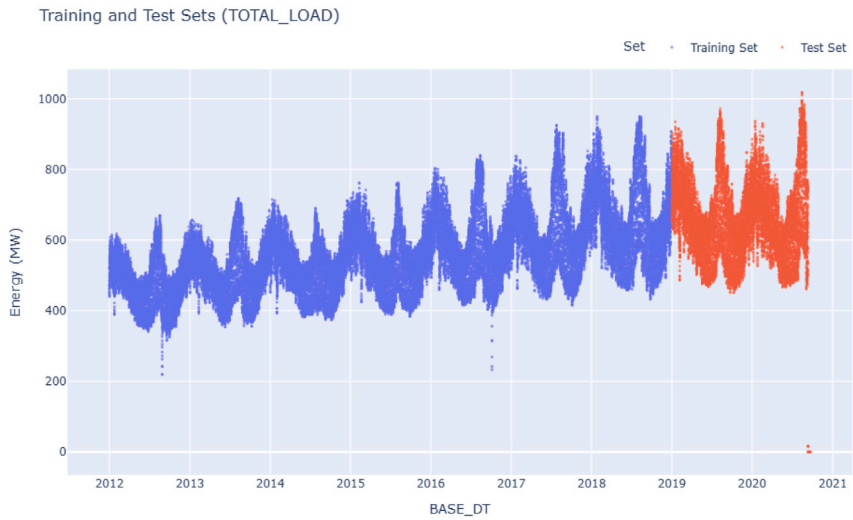


Fig. 8. Train–test split of the TOTAL-LOAD variable. Training covers 2012–2018 (blue), while 2019–2020 (red) is held out for evaluation.

Table 5
Outlines the key hyperparameters and configuration settings for both the Prophet and Temporal Fusion Transformer (TFT) models used in this study.

Prophet model parameters			Temporal Fusion Transformer (TFT) parameters		
Parameter	Value	Description	Parameter	Value	Description
Seasonality Mode	multiplicative	Seasonal effects scale with trend	Learning Rate	0.003	Step size for weight updates
Changepoint Prior Scale	0.1	Controls trend flexibility	Hidden Size	32	Size of hidden layers
Seasonality Prior Scale	10	Flexibility of seasonal components	Attention Heads	2	Number of attention heads
Daily Seasonality	True	Model daily seasonality	Dropout	0.2	Regularization rate
Weekly Seasonality	True	Model weekly seasonality	Hidden Continuous Size	16	Hidden units for continuous features
Yearly Seasonality	True	Model yearly seasonality	Max Prediction Length	24	Forecast horizon (1–24 h)
Monthly Period	30.5	Monthly seasonal period	Max Encoder Length	30	Input sequence length
Monthly Fourier Order	5	Fourier order for monthly seasonality	Quantiles	[0.1, 0.5, 0.9]	Quantile levels for forecasts
Hourly Period	24	Hourly seasonal period	Output Size	3	Quantiles per horizon
Hourly Fourier Order	10	Fourier order for hourly seasonality	Loss Function	QuantileLoss()	Distributional forecasting loss
lag_1	Yes	Add 1-h lag regressor	Log Interval	10	Training steps between logs
lag_{24}	Yes	Add 24-h lag regressor	Plateau Patience	5	Epochs before LR reduction
Weather Features	Yes	Temperature, wind, humidity, etc.	–	–	–

Table 6
Average seasonal performance across Winter, Spring, Summer, and Fall. Prophet–TFT achieves the lowest errors on both point and probabilistic metrics, outperforming classical (e.g., GAMs) and hybrid deep learning baselines.

Hybrid model	Avg. MAE	Avg. RMSE	Avg. MAPE (%)	Avg. CRPS	Avg. Pinball
FFNN	14.7	19.25	2.18	1.55	1.63
CNN–LSTM	14.2	18.75	2.12	1.46	1.53
XGBoost–LSTM	14.0	18.55	2.06	1.47	1.55
GAMs	13.8	18.25	2.04	1.45	1.53
EMD–LSTM	13.2	18.05	2.02	1.43	1.51
Prophet–LSTM	13.7	18.15	1.90	1.44	1.51
Prophet–TFT	**8.4**	**11.25**	**1.43**	**1.40**	**1.48**

5. Experimental results

We focus on short-term, hour-ahead multi-step forecasting, where each model produces 1–24 h forecasts at every rolling origin. Although the prediction horizon is short-term, evaluation is carried out over a full annual test period (2019), so that monthly and seasonal error summaries reflect the aggregation of repeated short-term forecasts rather than a mid-term horizon. Both point and probabilistic accuracy are assessed: point metrics (MAE, RMSE, MAPE) are computed from the median (P50) forecast, while probabilistic metrics (CRPS, Pinball loss, coverage, sharpness) are computed from the full predictive distribution.

Table 6 reports the average seasonal results. The proposed Prophet–TFT achieves the best overall performance with Avg. MAE = 8.4, RMSE = 11.25, and MAPE = 1.43%, alongside the lowest Avg. CRPS = 1.40 and Pinball = 1.48. Compared to the strongest baselines, Prophet–TFT reduces MAE/RMSE by ≈36–38% relative to EMD–LSTM, and lowers MAPE by ≈25% relative to Prophet–LSTM. This demonstrates that the accuracy gains are consistent across both error magnitude and percentage-based measures.

Classical models such as Feedforward Neural Network (FFNN) and GAMs remain competitive but trail significantly: FFNN has the highest MAE (14.7) and RMSE (19.25), showing limitations in capturing nonlinearities. GAMs perform slightly better (MAE 13.8, RMSE 18.25) by modeling seasonal structure, but still underperform compared to deep learning hybrids. CNN–LSTM and XGBoost–LSTM strike a balance, achieving moderate MAE/RMSE reductions but weaker MAPE compared to Prophet–LSTM. EMD–LSTM, by combining decomposition with LSTM, is the strongest baseline on RMSE but still lags behind Prophet–TFT by a large margin. Prophet–LSTM, on the other hand, excels in capturing holiday and seasonal components, achieving the best baseline MAPE (1.90%) but weaker RMSE compared to EMD–LSTM.

Tables 7 and 8 break down seasonal performance. Prophet–TFT consistently achieves the lowest MAE/RMSE across all four seasons: Winter

Table 7
Winter and Spring performance. Prophet–TFT consistently attains the lowest MAE/RMSE; CNN–LSTM and Prophet–LSTM are competitive on select probabilistic/percentage metrics.

Hybrid model	Winter					Spring				
	MAE	RMSE	MAPE (%)	CRPS	Pinball	MAE	RMSE	MAPE (%)	CRPS	Pinball
FFNN	14.8	19.3	2.10	1.54	1.63	15.0	19.6	2.20	1.55	1.64
CNN–LSTM	14.2	18.8	2.05	**1.41**	**1.48**	14.5	19.0	2.12	1.51	1.59
XGBoost–LSTM	13.9	18.5	1.97	1.46	1.54	14.0	18.6	2.00	1.48	1.56
GAMs	13.7	18.2	1.92	1.44	1.52	13.8	18.2	2.01	1.45	1.54
EMD–LSTM	12.0	18.0	1.90	1.42	1.51	13.6	18.0	1.98	1.44	1.52
Prophet–LSTM	13.6	18.1	1.94	1.43	1.50	13.7	18.1	**1.25**	1.43	1.51
Prophet–TFT	**8.9**	**11.8**	**1.28**	1.40	1.47	**8.1**	**10.7**	1.93	**1.36**	**1.46**

Table 8
Summer and Fall performance. Prophet–TFT maintains the lowest MAE/RMSE and achieves the best MAPE in both seasons; CNN–LSTM and EMD–LSTM remain competitive on select probabilistic metrics.

Hybrid model	Summer					Fall				
	MAE	RMSE	MAPE (%)	CRPS	Pinball	MAE	RMSE	MAPE (%)	CRPS	Pinball
FFNN	14.3	18.9	2.18	1.52	1.60	14.7	19.2	2.25	1.57	1.65
CNN–LSTM	13.8	18.4	2.10	**1.39**	**1.45**	14.3	18.8	2.20	1.53	1.61
XGBoost–LSTM	14.0	18.6	2.11	1.45	1.53	14.0	18.5	2.16	1.50	1.58
GAMs	13.9	18.3	2.08	1.44	1.51	13.8	18.3	2.13	1.47	1.55
EMD–LSTM	13.7	18.2	2.07	1.43	1.50	13.5	18.0	2.12	**1.42**	**1.49**
Prophet–LSTM	13.9	18.4	2.09	1.44	1.51	13.6	18.1	2.14	1.44	1.51
Prophet–TFT	**8.2**	**11.0**	**1.22**	1.42	1.49	**8.4**	**11.5**	**1.27**	1.43	1.50

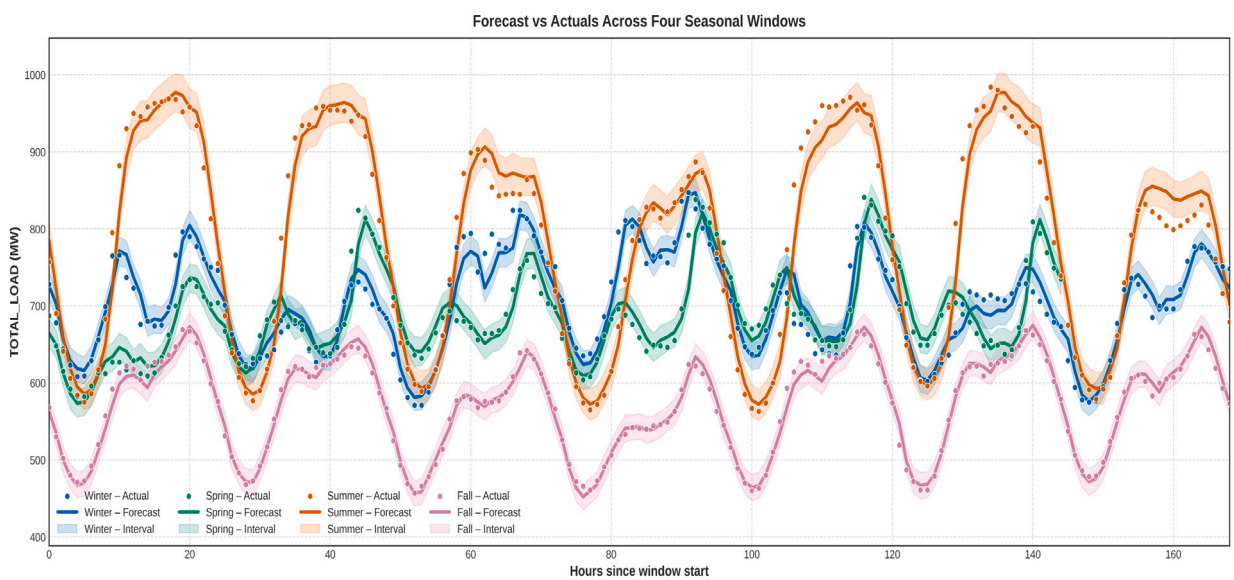


Fig. 9. Seasonal forecasts from Prophet–TFT. Colors encode seasons (Winter blue, Spring green, Summer orange, Fall purple). The model tracks daily peaks and maintains tight, well-calibrated uncertainty bands.

(8.9/11.8), Spring (8.1/10.7), Summer (8.2/11.0), and Fall (8.4/11.5). This robustness highlights its ability to handle both high-load heating and cooling periods. Prophet–LSTM achieves the lowest Spring MAPE (1.25%), illustrating its strength in capturing seasonal-level percentage accuracy, while CNN–LSTM shows competitive CRPS/Pinball in Winter. EMD–LSTM provides balanced performance but is consistently outperformed by Prophet–TFT. GAMs deliver stable but higher errors across all seasons, confirming their interpretability but limited competitiveness in highly nonlinear, weather-sensitive regimes. As shown in Fig. 9, the proposed Prophet–TFT model not only captures the regular daily cycles across seasons but also adapts to sudden demand spikes, particularly during holiday and peak-load periods. The forecast plots demonstrate that the model follows observed peaks closely while keeping predictive intervals tight and well-calibrated, supporting our claim of robustness under both routine and irregular demand conditions.

Calibration results in Table 9 further strengthen these findings. Prophet–TFT attains the sharpest intervals (19.6 kW) while maintaining near-nominal coverage (P50 = 50.2%, P90 = 91.1%). Prophet–LSTM and EMD–LSTM also yield strong calibration, but with slightly wider intervals. Feedforward Neural Network (FFNN) and GAMs provide acceptable coverage but much lower sharpness, highlighting their weaker uncertainty modeling. CNN–LSTM and XGBoost–LSTM improve interval sharpness relative to FFNN but remain less competitive than Prophet–TFT.

Finally, Table 10 compares non-holiday and holiday conditions. As expected, all models degrade on holidays due to irregular demand spikes. Feedforward Neural Network (FFNN) shows the steepest deterioration, confirming its sensitivity to shocks. GAMs are more stable but still exhibit increased MAE and RMSE. CNN–LSTM and

Table 9
Calibration quality. Prophet–TFT delivers near-nominal coverage (P50/P90) with the sharpest predictive intervals, indicating reliable uncertainty quantification.

Models	P50 coverage (%)	P90 coverage (%)	Sharpness (kW)
FFNN	47.8	88.7	22.3
CNN–LSTM	48.4	89.5	21.7
XGBoost–LSTM	48.9	89.9	21.1
GAMs	49.2	90.2	20.6
EMD–LSTM	49.4	90.4	20.3
Prophet–LSTM	49.5	90.6	20.0
Prophet–TFT	**50.2**	**91.1**	**19.6**

XGBoost–LSTM improve over these baselines but do not provide robustness across holidays. Prophet–LSTM and EMD–LSTM perform competitively, but Prophet–TFT demonstrates the smallest performance drop (MAE 8.3 → 9.7, RMSE 11.2 → 12.9) and retains the best CRPS/Pinball in both regimes. This indicates that combining Prophet's holiday-awareness with TFT's feature interactions delivers superior stability under irregular, high-variance conditions.

5.1. Model interpretability

The Temporal Fusion Transformer (TFT) includes an intrinsic interpretability mechanism via its variable selection networks, which assign input-dependent gating weights to features at each time step (Zhou et al., 2024). These weights reflect the relative importance of different covariates during both the encoding (historical context) and decoding (forecasting) phases. While these importance scores are not causal in the strict sense emphasized in the forecasting literature, they provide valuable insight into how the model distributes attention across inputs (Koya and Roy, 2023).

Our results show that autoregressive load features dominate the forecasts Fig. 10. Specifically, the previous hour's load lag1 was the most influential feature in both encoder and decoder phases, followed by the current load (`TOTAL_LOAD`) and the previous day's load lag24. Temporal indicators such as `hour` and `time_idx` were also highly ranked, consistent with the importance of diurnal cycles and daily recurrence patterns in short-term load forecasting. Weather variables contributed moderately, Table 11 shows the most influential weather-related features, while Prophet-derived components (trend and seasonality) showed minimal importance. This indicates that the TFT was able to internalize temporal structures directly from the data, reducing the added value of explicit decomposition.

To validate and contextualize the TFT's internal importance scores, we performed a SHAP analysis in Fig. 11, which provides a model-agnostic explanation by quantifying each feature's marginal contribution to predictions. The SHAP results corroborate the TFT findings: short-term temporal features (`time_idx`, `hour`, and autoregressive lags) had the greatest impact on model outputs, while weather-related covariates (e.g., solar irradiance and wind speed at Gosan, Jeju, and Seogwipo) exhibited secondary contributions. Prophet-derived components again showed very limited effect. The consistency between TFT variable selection and SHAP analysis strengthens our interpretability claims. Both perspectives highlight that forecasts are overwhelmingly driven by recent load history and time-based features, with weather serving as a secondary contextual signal. At the same time, the marginal importance of decomposed Prophet trend and seasonality confirms that these inputs were not essential, but their inclusion offered a robustness check verifying the TFT's capacity to capture such patterns internally.

In summary, the combined use of TFT's built-in variable importance and SHAP analysis provides both model-intrinsic and model-agnostic interpretability. While neither offers causal guarantees, together they demonstrate that the model's forecasting performance relies primarily on autoregressive load and temporal covariates, with limited added benefit from explicit decomposition. This directly addresses the concern that our interpretability claims may be overstated, showing instead that our analysis provides consistent and reliable insights into the model's decision-making process.

6. Discussion

This study confirms the effectiveness of the proposed Prophet–TFT hybrid framework for short-term electricity load forecasting on Jeju Island. By integrating Prophet's statistical decomposition with TFT's deep sequence modeling, the hybrid consistently outperformed both classical models (e.g., ARIMA, GAMs) and advanced hybrids (e.g., CNN–LSTM, Prophet–LSTM, EMD–LSTM). Across the 2019–2020 evaluation period, the hybrid achieved the lowest average errors (MAE = 8.4 MW, RMSE = 11.25 MW, MAPE = 1.43%), improving upon EMD–LSTM by 36%–38% in magnitude-based errors and upon Prophet–LSTM by ~25% in percentage accuracy. Probabilistic evaluation further highlighted its strengths, with the lowest CRPS (1.40) and Pinball loss (1.48), together with sharp yet well-calibrated prediction intervals (19.6 kW width; P50 ≈ 50.2%, P90 ≈ 91.1%). Seasonal and regime analyses reinforced the model's robustness. Prophet–TFT maintained leading accuracy across Winter, Spring, Summer, and Fall, and its performance degradation during holidays was minimal (MAE increase of only 1.4 MW, compared with >2.5 MW for CNN–LSTM and FFNN). These results demonstrate the framework's capacity to generalize across routine and irregular demand regimes, a critical advantage for operational forecasting. Interpretability analyses also validated the hybrid's transparency. Both the TFT's variable selection networks and SHAP revealed that recent load history (lag1, lag24) and temporal indicators (hour, day of week, season) were the dominant predictors, while weather variables (solar irradiance, wind speed, temperature) provided complementary signals during extreme conditions. Prophet-derived components contributed less directly to predictive power, but their inclusion strengthened robustness and ensured the model's seasonal and holiday awareness. From a practical perspective, these findings indicate that the Prophet–TFT framework not only improves forecast accuracy but also provides well-calibrated uncertainty estimates and interpretable feature attributions key requirements for grid operators managing demand-side planning, renewable integration, and system stability. Nevertheless, limitations remain. The framework's higher computational cost compared with lightweight baselines may constrain real-time or resource-limited deployment. Its reliance on historical weather–load relationships also introduces vulnerability to concept drift under climate change and evolving usage patterns. Additionally, the outlier removal strategy, while stabilizing, may exclude rare but operationally important events. Future work should therefore explore adaptive retraining for drift detection, more robust outlier handling methods, and lightweight model compression for edge deployment. Extending the framework to multi-region and multi-energy systems would further validate its generalizability.

7. Conclusion

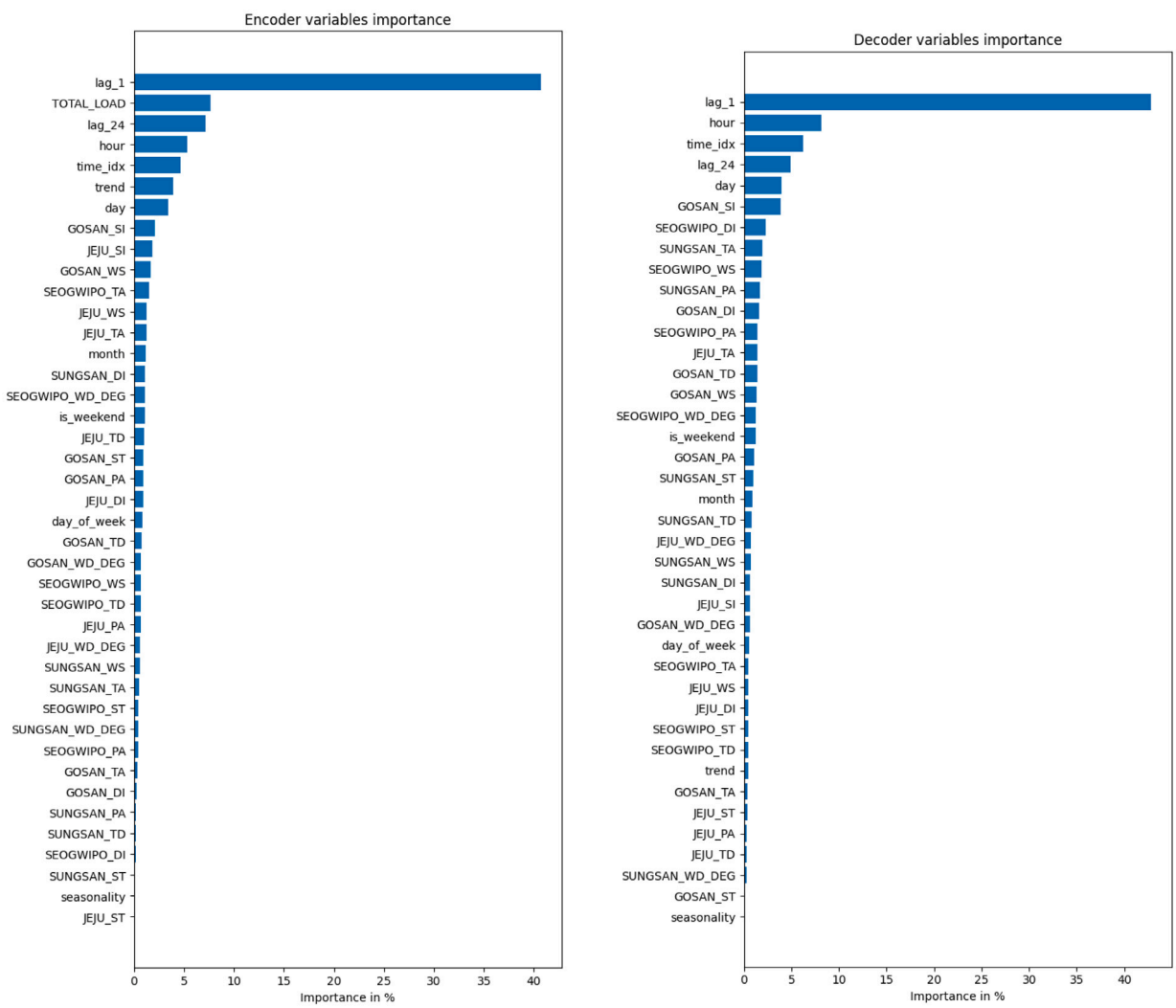
This study proposed a hybrid Prophet–Temporal Fusion Transformer (TFT) framework for short-term electricity load forecasting, combining the interpretability of statistical decomposition with the representational power of deep learning. By integrating Prophet's trend, seasonal, and holiday components with TFT's ability to capture nonlinear dynamics and multi-scale dependencies, the model consistently outperformed classical statistical methods and state-of-the-art hybrids. Experiments on nine years of Jeju Island's hourly load and meteorological data demonstrated significant gains across all evaluation metrics (MAE, RMSE, MAPE, CRPS, and pinball loss), with robust accuracy even under holidays and extreme weather conditions. In addition to predictive improvements, the framework enhanced interpretability: attention-based feature importance and SHAP analysis revealed that recent load history and temporal indicators were dominant drivers, with weather

Table 10
Holiday vs. non-holiday performance. All models incur higher errors on holidays due to irregular demand spikes, whereas Prophet–TFT shows the smallest degradation and the best calibration in both regimes.

Hybrid model	MAE		RMSE		MAPE (%)		CRPS		Pinball	
	N-Holiday	Holiday	N-Holiday	Holiday	N-Holiday	Holiday	N-Holiday	Holiday	N-Holiday	Holiday
FFNN	14.5	17.2	19.0	22.4	2.1	2.6	1.55	1.72	1.63	1.80
CNN–LSTM	13.9	16.1	18.5	21.0	2.0	2.4	1.46	1.60	1.53	1.69
XGBoost–LSTM	13.7	15.7	18.3	20.6	1.9	2.3	1.47	1.58	1.55	1.65
GAMs	13.4	15.5	18.0	20.3	1.9	2.2	1.45	1.56	1.53	1.62
EMD–LSTM	13.2	15.2	17.9	20.1	1.8	2.2	1.43	1.54	1.51	1.60
Prophet–LSTM	13.3	15.4	18.1	20.4	1.8	2.3	1.44	1.55	1.51	1.62
Prophet–TFT	**8.3**	**9.7**	**11.2**	**12.9**	**1.2**	**1.5**	**1.40**	**1.47**	**1.48**	**1.55**

Table 11
Top five weather-related features ranked by their combined importance in the forecasting model. These features capture key environmental influences on electricity demand.

Rank	Feature name	Description	Combined importance (%)
1	GOSAN_SI	Solar irradiance at Gosan	5.3
2	GOSAN_WS	Wind speed at Gosan	2.5
3	JEJU_TA	Air temperature at Jeju	2.2
4	SEOGWIPO_WS	Wind speed at Seogwipo	2.0
5	JEJU_SI	Solar irradiance at Jeju	2.0



(a) Encoder variables importance. (b) Decoder variables importance.

Fig. 10. Illustrate the relative importance of input variables for the Temporal Fusion Transformer (TFT) model during the encoding and decoding phases of time series forecasting.

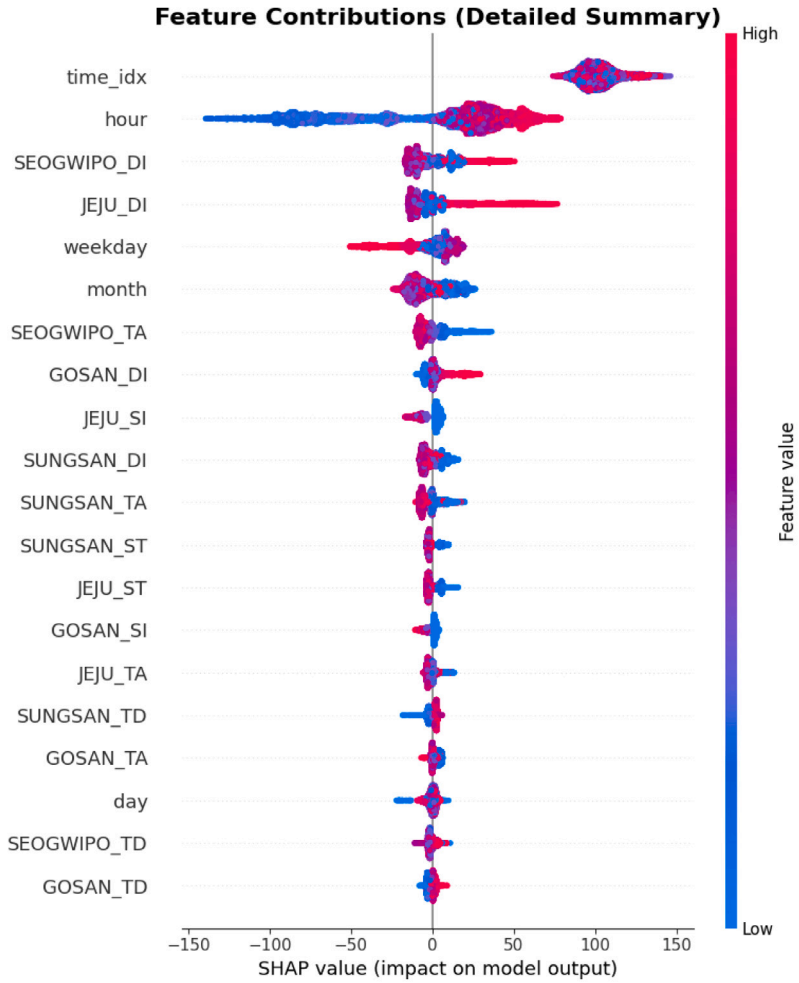


Fig. 11. SHAP summary plot showing feature-level contributions to model output. Features are ranked by overall impact, with color indicating feature value (high to low).

covariates such as solar irradiance, wind speed, and air temperature providing complementary signals. This transparency supports operator trust and delivers actionable insights for grid management. Practically, the model enables more reliable demand forecasting for system planning, renewable integration, and operational stability. While the hybrid's complexity is higher than lightweight alternatives, its accuracy and explainability justify its use in mission-critical forecasting. Future work will extend the framework to multi-region and multi-energy settings, introduce adaptive retraining to address data and concept drift, and develop lightweight variants suitable for edge deployment. We also plan to replace naïve outlier deletion with two regression-based, time-series-aware methods: regularized calendar and weather-driven quantile regression to generate robust conditional bands and flag or adjust extremes, and targeted down-weighting of transient spikes during re-fitting to preserve sequence continuity.

CRediT authorship contribution statement

Nazrul Amin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yong-Woon Kim:** Writing – review & editing, Visualization, Validation, Conceptualization. **Chulung Kang:** Writing – review & editing. **Yung-Cheol Byun:** Validation, Methodology, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors used GPT-4 to improve the language and readability while preparing this work. Following the use of this tool or service, the author or authors took full responsibility for the publication's content and reviewed and edited it as necessary.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2024-00405278), This research was also supported by the Regional Innovation System & Education (RISE) program through the Jeju RISE center, funded by the Ministry of Education (MOE), South Korea and the Jeju Special Self-Governing Province, Republic of Korea (2025-RISE-17-001).

Data availability

Data will be made available on request.

References

Abumohsen, M., Owda, A.Y., Owda, M., 2023. Electrical load forecasting using lstm, gru, and rnn algorithms. Energies 16 (5), 2283.

Aditya, T., Jaipuria, S., Kumar Dadabada, P., 2025. A review of methods for long-term electric load forecasting. J. Forecast. 44 (4), 1403–1423.

ArunKumar, K., Kalaga, D.V., Kumar, C.M.S., Kawaji, M., Brenza, T.M., 2022. Comparative analysis of gated recurrent units (gru), long short-term memory (lstm) cells, autoregressive integrated moving average (arima), seasonal autoregressive integrated moving average (sarima) for forecasting covid-19 trends. Alex. Eng. J. 61 (10), 7585–7603.

Aslam, S., Herodotou, H., Mohsin, S.M., Javaid, N., Ashraf, N., Aslam, S., 2021. A survey on deep learning methods for power load and renewable energy forecasting in smart microgrids. Renew. Sustain. Energy Rev. 144, 110992.

Azeem, A., Ismail, I., Jameel, S.M., Harindran, V.R., 2021. Electrical load forecasting models for different generation modalities: A review. IEEE Access 9, 142239–142263.

Bashir, T., Haoyong, C., Tahir, M.F., Liqiang, Z., 2022. Short term electricity load forecasting using hybrid prophet-lstm model optimized by bpnn. Energy Rep. 8, 1678–1686.

Bassey, K.E., 2023. Hybrid renewable energy systems modeling. Eng. Sci. Technol. J. 4 (6), 571–588.

Chaturvedi, S., Rajasekar, E., Natarajan, S., McCullen, N., 2022. A comparative assessment of sarima, lstm rnn and fb prophet models to forecast total and peak monthly energy demand for India. Energy Policy 168, 113097.

Chen, Z., Ma, M., Li, T., Wang, H., Li, C., 2023. Long sequence time-series forecasting with deep learning: A survey. Inf. Fusion 97, 101819.

Coutinho, E.R., Madeira, J.G., Borges, D.G., Springer, M.V., de Oliveira, E.M., Coutinho, A.L., 2025. Multi-step forecasting of meteorological time series using cnn-lstm with decomposition methods. Water Resour. Manag. 1–26.

Dai, T.-Y., Niyogi, D., Nagy, Z., 2025. Citytft: A temporal fusion transformer-based surrogate model for urban building energy modeling. Appl. Energy 389, 125712.

Du, J., Hu, M., Zhang, W., 2020. Missing data problem in the monitoring system: A review. IEEE Sensors J. 20 (23), 13984–13998.

Fan, G.-F., Yu, M., Dong, S.-Q., Yeh, Y.-H., Hong, W.-C., 2021. Forecasting short-term electricity load using hybrid support vector regression with grey catastrophe and random forest modeling. Util. Policy 73, 101294.

Fan, J., Zhuang, W., Xia, M., Fang, W., Liu, J., 2024. Optimizing attention in a transformer for multihorizon, multienergy load forecasting in integrated energy systems. IEEE Trans. Ind. Informatics 20 (8), 10238–10248.

Gaillard, P., Goude, Y., Nedellec, R., 2016. Additive models and robust aggregation for gefcom2014 probabilistic electric load and electricity price forecasting. Int. J. Forecast. 32 (3), 1038–1050.

Gong, J., Qu, Z., Zhu, Z., Xu, H., Yang, Q., 2025. Ensemble models of tcn-lstm-lightgbm based on ensemble learning methods for short-term electrical load forecasting. Energy 318, 134757.

Hamed, E.A.-R., Salem, M.A.-M., Badr, N.L., Tolba, M.F., 2023. An efficient combination of convolutional neural network and lightgbm algorithm for lung cancer histopathology classification. Diagnostics 13 (15), 2469.

Hammad, M.A., Jereb, B., Rosi, B., Dragan, D., 2020. Methods and models for electric load forecasting: a comprehensive review, logistics, supply chain. Sustain. Glob. Challenges 11 (1), 51–76.

Jain, A., Gupta, S., 2024. Evaluation of electrical load demand forecasting using various machine learning algorithms. Front. Energy Res. 12, 1408119.

Khamma, T.R., Zhang, Y., Guerrier, S., Boubekri, M., 2020. Generalized additive models: An efficient method for short-term energy prediction in office buildings. Energy 213, 118834.

Koehler, A.B., Snyder, R.D., Ord, J.K., 2001. Forecasting models and prediction intervals for the multiplicative holt–winters method. Int. J. Forecast. 17 (2), 269–286.

Koprinska, I., Wu, D., Wang, Z., 2018. Convolutional neural networks for energy time series forecasting. In: 2018 International Joint Conference on Neural Networks. IJCNN, IEEE, pp. 1–8.

Koya, S., Roy, T., 2023. Temporal fusion transformers for streamflow prediction: Value of combining attention with recurrence. arxiv. org.

Koya, S.R., Roy, T., 2024. Temporal fusion transformers for streamflow prediction: Value of combining attention with recurrence. J. Hydrol. 637, 131301.

Kuster, C., Rezgui, Y., Mourshed, M., 2017. Electrical load forecasting models: A critical systematic review. Sustain. Cities Soc. 35, 257–270.

Li, D., Tan, Y., Zhang, Y., Miao, S., He, S., 2023. Probabilistic forecasting method for mid-term hourly load time series based on an improved temporal fusion transformer model. Int. J. Electr. Power Energy Syst. 146, 108743.

Li, X., Wang, Z., Yang, C., Bozkurt, A., 2024. An advanced framework for net electricity consumption prediction: Incorporating novel machine learning models and optimization algorithms. Energy 296, 131259.

Lim, B., Arık, S.Ö., Loeff, N., Pfister, T., 2021. Temporal fusion transformers for interpretable multi-horizon time series forecasting. Int. J. Forecast. 37 (4), 1748–1764.

Liu, Y., Pu, H., Sun, D.-W., 2021. Efficient extraction of deep image features using convolutional neural network (cnn) for applications in detecting and analysing complex food matrices. Trends Food Sci. Technol. 113, 193–204.

Ma, P., Ren, J., Sun, G., Zhao, H., Jia, X., Yan, Y., Zabalza, J., 2023. Multiscale superpixelwise prophet model for noise-robust feature extraction in hyperspectral images. IEEE Trans. Geosci. Remote Sens. 61, 1–12.

Mansoor, M., Grimaccia, F., Leva, S., Mussetta, M., 2021. Comparison of echo state network and feed-forward neural networks in electrical load forecasting for demand response programs. Math. Comput. Simulation 184, 282–293.

Mathivha, F., Sigauke, C., Chikoore, H., Odiyo, J., 2020. Short-term and medium-term drought forecasting using generalized additive models. Sustainability 12 (10), 4006.

Mounir, N., Ouadi, H., Jrhilifa, I., 2023. Short-term electric load forecasting using an emd-bi-lstm approach for smart grid energy management system. Energy Build. 288, 113022.

Naima, S., Su, N., Kevin, C., Victor, G., Prem, C., 2023. Forecasting container freight rates using the prophet forecasting method. Transp. Policy 133, 86–107.

Nazir, A., Shaikh, A.K., Shah, A.S., Khalil, A., 2023. Forecasting energy consumption demand of customers in smart grid using temporal fusion transformer (tft). Results Eng. 17, 100888.

Nielsen, D., 2016. Tree Boosting with Xgboost-Why Does Xgboost Win Every Machine Learning Competition? (Master's thesis). NTNU.

Nti, I.K., Teimeh, M., Nyarko-Boateng, O., Adekoya, A.F., 2020. Electricity load forecasting: a systematic review. J. Electr. Syst. Inf. Technol. 7 (1), 13.

Peng, H., Jiang, B., Mao, Z., Liu, S., 2023. Local enhancing transformer with temporal convolutional attention mechanism for bearings remaining useful life prediction. IEEE Trans. Instrum. Meas. 72, 1–12.

Qiu, Y., Wang, S., Zhang, S., Xu, J., 2023. A novel gated dual convolutional neural network model with autoregressive method and attention mechanism for probabilistic load forecasting. Appl. Intell. 53 (17), 20256–20271.

Rafferty, G., 2023. Forecasting Time Series Data with Prophet: Build, Improve, and Optimize Time Series Forecasting Models using Meta's Advanced Forecasting Tool. Packt Publishing Ltd.

Rafi, S.H., Deeba, S.R., Hossain, E., et al., 2021. A short-term load forecasting method using integrated cnn and lstm network. IEEE Access 9, 32436–32448.

Shehadeh, A., Alshboul, O., Al Mamlook, R.E., Hamedat, O., 2021. Machine learning models for predicting the residual value of heavy construction equipment: An evaluation of modified decision tree, lightgbm, and xgboost regression. Autom. Constr. 129, 103827.

Song, Xingchen, Wang, Tongxi, Xu, Ziming, Shi, Zhan, 2022a. Multistep forecasting method of short-term power load based on vmd-prophet-seq2seq.. IAENG Internat. J. Comput. Sci. 49 (3).

Song, J., Zhang, L., Jiang, Q., Ma, Y., Zhang, X., Xue, G., Shen, X., Wu, X., 2022b. Estimate the daily consumption of natural gas in district heating system based on a hybrid seasonal decomposition and temporal convolutional network model. Appl. Energy 309, 118444.

Stefenon, S.F., Seman, L.O., Mariani, V.C., Coelho, L.d.S., 2023. Aggregating prophet and seasonal trend decomposition for time series forecasting of italian electricity spot prices. Energies 16 (3), 1371.

Stienecker, M., Hagemeier, A., 2023. Developing feedforward neural networks as benchmark for load forecasting: Methodology presentation and application to hospital heat load forecasting. Energies 16 (4).

Szostek, Kamil, Mazur, Damian, Drałus, Grzegorz, Kusznier, Jacek, 2024. Analysis of the effectiveness of arima, sarima, and svr models in time series forecasting: a case study of wind farm energy production.. Energies (19961073) 17 (19).

Tarmanini, C., Sarma, N., Gezegin, C., Ozgonenel, O., 2023. Short term load forecasting based on arima and ann approaches. Energy Rep. 9, 550–557.

Taylor, S.J., Letham, B., 2018. Forecasting at scale. Amer. Statist. 72 (1), 37–45.

Toharudin, T., Pontoh, R.S., Caraka, R.E., Zahroh, S., Lee, Y., Chen, R.C., 2023. Employing long short-term memory and facebook prophet model in air temperature forecasting. Comm. Statist. Simulation Comput. 52 (2), 279–290.

Valipour, M., 2015. Long-term runoff study using sarima and arima models in the united states. Meteorol. Appl. 22 (3), 592–598.

Wan, A., Chang, Q., Khalil, A.-B., He, J., 2023. Short-term power load forecasting for combined heat and power using cnn-lstm enhanced by attention mechanism. Energy 282, 128274.

Wang, W., He, N., Chen, M., Jia, P., 2024a. Freight rate index forecasting with prophet model based on multi-dimensional significant events. Expert Syst. Appl. 249, 123451.

Wang, Y., Mao, M., Chang, L., 2024b. Multi-time scales prediction of aggregated schedulable capacity of electric vehicle fleets based on enhanced prophet-lgbm algorithm. Appl. Energy 374, 124018.

Wen, Q, Zhou, T, Zhang, C, Chen, W, Ma, Z, Yan, J, 2022. Transformers in time series: a survey. arXiv preprint arXiv:220207125.

Weron, R., 2006. Modeling and Forecasting Electricity Loads and Prices: A Statistical Approach. John Wiley & Sons.

Wu, B., Wang, L., Zeng, Y.-R., 2022. Interpretable wind speed prediction with multivariate time series and temporal fusion transformers. Energy 252, 123990.

Xue, J., Hu, X., Chen, H., Zhou, G., 2022. Research on lstm-xgboost integrated model of photovoltaic power forecasting system. In: 2022 14th International Conference on Intelligent Human-Machine Systems and Cybernetics. IHMSC, IEEE, pp. 22–25.

Yazici, I., Beyca, O.F., Delen, D., 2022. Deep-learning-based short-term electricity load forecasting: A real case application. Eng. Appl. Artif. Intell. 109, 104645.

Zha, W., Liu, Y., Wan, Y., Luo, R., Li, D., Yang, S., Xu, Y., 2022. Forecasting monthly gas field production based on the cnn-lstm model. Energy 260, 124889.

Zhang, Y.-M., Wang, H., 2023. Multi-head attention-based probabilistic cnn-bilstm for day-ahead wind speed forecasting. Energy 278, 127865.
Zhang, J., Wei, Y.-M., Li, D., Tan, Z., Zhou, J., 2018. Short term electricity load forecasting using a hybrid model. Energy 158, 774–781.
Zhao, D., Zhang, R., Zhang, H., He, S., 2022. Prediction of global omicron pandemic using arima, mlr, and prophet models. Sci. Rep. 12 (1), 18138.
Zhou, H., Zheng, P., Dong, J., Liu, J., Nakanishi, Y., 2024. Interpretable feature selection and deep learning for short-term probabilistic pv power forecasting in buildings using local monitoring data. Appl. Energy 376, 124271.
Ziel, F., Steinert, R., 2018. Probabilistic mid-and long-term electricity price forecasting. Renew. Sustain. Energy Rev. 94, 251–266.